

Timeless Timing Attacks: Exploiting Concurrency to Leak Secrets over Remote Connections

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Timeless Timing Attacks: Exploiting Concurrency to Leak Secrets over Remote Connections

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To perform successful remote timing attacks, an adversary typically collects a series of network timing measurements and subsequently performs statistical analysis to reveal a difference in execution time. The number of measurements that must be obtained largely depends on the amount of jitter that the requests and responses are subjected to. In remote timing attacks, a significant source of jitter is the network path between the adversary and the targeted server, making it practically infeasible to successfully exploit timing side-channels that exhibit only a small difference in execution time.

In this paper, we introduce a conceptually novel type of timing attack that leverages the coalescing of packets by network protocols and concurrent handling of requests by applications. These concurrency-based timing attacks infer a relative timing difference by analyzing the order in which responses are returned, and thus do not rely on any absolute timing information. We show how these attacks result in a 100-fold improvement over typical timing attacks performed over the Internet, and can accurately detect timing differences as small as 100ns, similar to attacks launched on a local system. We describe how these timing attacks can be successfully deployed against HTTP/2 web servers, Tor onion services, and EAP-pwd, a popular Wi-Fi authentication method.

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